Viscous Damping Coefficient of a moving object

BY

THU TA ZAW

Measure the viscous damping coefficient of something moving.

Conduct a mini-experiment at the lowest possible cost.

Verify that you actually measured the viscous damping, not friction.

Note: Please show not only the experimental results, but also the idea (how to measure), measurement theory, theoretical resolution, accuracy, cost, evaluation/discussion etc.

In this assignment, I tried to measure the viscous damping coefficient c, of a stack of coin moving vertically in water.

Five one-yen coins were stacked together with tape, which behaved like one rigid cylinder. The 1-L plastic container with transparent side was filled with water. A rule was attached to the side of the container to provide a scale. A smartphone camera with 60 fps was used to record the falling object.

The weight and size of the coins were obtained from the official published documents and used them as a reference.

The stack of coins is assumed as a rigid body moving straight down.

Free body diagram of the coin stack in the water

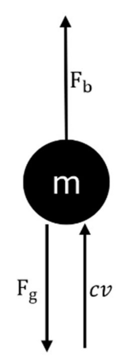

Fig.1 Free body diagram of the coin stack in the water

Forces acting on the coins stacked are:

Gravity: $F_g = mg$

Buoyancy: $F_b = \rho_f Vg$ (upward)

Viscous Drag: $F_d = -cv$ (upward, proportional to velocity)

Where:

ρ_f = water density

V = volume of coin stack

M = mass of coin stacks

V = downward velocity.

Equation of motion:

$$m\frac{dv}{dt} = \mathrm{F_g} - \mathrm{F_b} - cv = mg - \rho_{\mathrm{f}}\mathrm{Vg} - \mathrm{cv}$$

Effective weight:

$$\mathrm{W_{eff}} = (m - \rho_{\mathrm{f}}\mathrm{V})g$$

Then:

$$m\frac{dv}{dt} = \mathrm{W_{eff}} - cv$$

This is a first-order linear ODE. Solve with $v(0) = 0$:

$$\mathrm{v(t)} = \mathrm{v_t}\,(1 - \mathrm{e}^{\frac{-\mathrm{t}}{\tau}})$$

where:

Terminal velocity:

$$\mathrm{v_t} = \frac{\mathrm{W_{eff}}}{c}$$

Time constant:

$$\tau = \frac{\mathrm{m}}{c}$$

So,

$$c = \frac{\mathrm{W_{eff}}}{\mathrm{v_t}}$$

Name	Value
Weight	1.0 g
Diameter	20.0 mm
Thickness	1.2 mm
Density	2.7 g/cm

Table. 1-yen coin data

The experiment was performed by using simple household items. Starting with, five one-yen coins were stacked with tape, and then a clear 1 L water bottle was filled with water, and a paper with measurement markers was taped to the surface of the bottle for the measurement. After the preparation, the video is recorded by the mobile phone, and the coin stack was dropped from the start of the water level to the bottom of the water bottle. The time taken by the coin stacked to travel from the top to the bottom of the water bottle is noted.

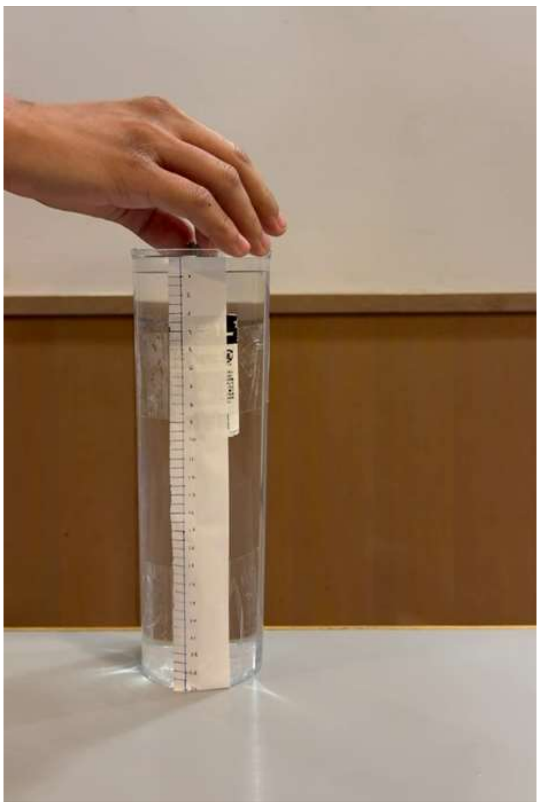

Fig.2 The starting point of the coin stacked before the release into the water bottle

Fig.3 The end point of the coin after travelling for 1.1s after releasing

According to the experiment, the coin stack travels 20.5 cm in 1.1s.

Distance: y = 0.205 m

Time: t = 1.1 s

I assumed the velocity that calculated from the experiment as an average velocity.

$$v_{avg} = \frac{y}{t} = \frac{0.205}{1.1} = 0.186\ ms^{-1}$$

I will assume that the coin stack quickly reaches terminal speed and then moves almost at v_t,

$$v_t \approx\ 0.186\ ms^{-1}$$

Where:

Mass: m = 0.005 kg

Volume: V = 1.88 x $10^{-6}\ m^3$

Water density: ρ_f = 1000 kgm^{-3}

Gravity: g = 9.81 ms^{-2}

Buoyant mass:

$$m_{water} = \rho_f V = 1.88 \, x \, 10^{-3} kg$$

Effective mass:

$$m_{eff} = m - m_{water} = 0.00312 \, kg$$

Effective weight:

$$W_{eff} = m_{eff} g = 0.00312 \, x \, 9.81 \approx 0.0306 \, N$$

$$c = \frac{\mathrm{W_{eff}}}{\mathrm{v_t}} \approx \frac{0.0306}{0.186} \approx 0.164 Nm^{-1}$$

$$c \approx 0.164 \, Nm^{-1}$$

$$\tau = \frac{\mathrm{m}}{c} \approx \frac{0.005}{0.164} \approx 0.030 \, s$$

$$\tau \approx 0.030 \, s$$

$$3\tau \approx 0.090 \, s$$

By analyzing the result, the coin stack will already arrive >95% of its terminal speed within $\sim 0.1 \, s$. Since the object had no solid-to-solid contact, there were no coulomb friction and the coin stack moved freely in the water. Moreover, the very short time constant showed that the coin stacked had constant speed shortly after the release. Since there was no solid friction, the measured damping arises from the viscous drag, not friction.

All the equipment used in this experiment had lowest possible cause. Although the possible measured from the video resolution and possibility of slight sideways drift of the coin stack, the experiment still demonstrates real measurable viscous behavior.

THU TA ZAW

2415121095